Supplementary Material for "FastDC: Fast KV Dimensionality Compression for Decompression-Free and Attention-Accelerated LLM Inference"

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Abstract

This document is the supplementary material for the SoCC '26 paper "FastDC: Fast KV Dimensionality Compression for Decompression-Free and Attention-Accelerated LLM Inference" [10]. It collects the extended results behind the paper's statements, in the order in which the paper cites them. Each section heading names the section of the paper that cites it. The models, datasets, metrics, and notation are those of the paper.

1 Basis-Invariant Comparison of Rotations (Paper §3.2)

For each head h in layer l, the paper computes a rotation matrix $R_r^{h,l}$ from random tokens and a rotation matrix $R_a^{h,l}$ from each dataset a, and it compares the two with a difference ratio (Table 3 of the paper). As a basis-invariant check, the last row of that table reports, per model, the largest principal angle between the subspaces that $R_r^{h,l}$ and $R_a^{h,l}$ retain at the 99% operating points, maximized over heads, layers, and datasets. An operating point is the set of compression rates that retains 99% of the uncompressed accuracy (paper §5.3). Table 1 gives this maximum per dataset for the two Llama-3.1 models, maximized over heads and layers only. The per-model maxima, 1.17° and 1.20°, match the last row of the paper's Table 3. The medians over heads, layers, and datasets are 0.15° for Mistral-v0.3 7B, 0.29° for Llama-3.1 8B, 0.27° for Phi-3 14B, 0.26° for Qwen3.5 27B, 0.23° for Yi 34B, and 0.28° for Llama-3.1 70B.

Table 1: Largest principal angle (degrees) between the subspaces retained by $R_r^{h,l}$ and $R_a^{h,l}$ at the 99% operating points, maximized over heads and layers, per dataset.

	Llama-3.1 8B	Llama-3.1 70B
IMDb	0.99	0.96
HumanEval	1.17	1.17
GSM8K	0.68	0.96
arXiv	1.20	1.05
Cocktail	0.83	0.81
RULER-128K	0.95	0.87
SCBench-MT	1.04	1.01
PG-19	1.16	0.93

Column norms of the weights by coordinate pair. The paper (§3.2) argues that whether a coordinate pair of Q and K is small for every token is decided by the two corresponding columns of W_Q and W_K. It also reports that the subspace discarded by R carries a median of 5% of the column energy of W_Q and 7% of that of W_K. The following measurement compares this data-derived subspace with a ranking of the weights themselves. For each KV head of Llama-3.1 70B, the 64 coordinate pairs of W_Q (summed over the query heads of the group) and of W_K are ranked by the pair norm $\sqrt{\|w_{2m}\|^2 + \|w_{2m+1}\|^2}$, where w_c is column c. Over heads and layers, the 32 smallest pairs (half of them) hold a median of 5% of the column energy of W_Q and 6% of that of W_K. The ratio between the largest and the smallest pair norm has a median of 13 for W_Q and 12 for W_K. The subspace that the singular value decomposition of the data after Rotary Positional Embedding (RoPE) discards at the 99% operating points carries 5% and 7% of the column energy, respectively, so the low-energy subspaces chosen by the weights and by the data carry the same share of column energy to within one percentage point.

2 Coefficient of Variation of the Maximum Singular Values (Paper §4.2)

The adaptive compression of the paper assigns each head its own compression rate because the maximum singular values differ across heads. Table 4 of the paper quantifies this difference per model as the range over the eight datasets of the coefficient of variation (CV) of the maximum singular values across heads, that is, the ratio of their standard deviation to their mean. Tables 2 and 3 give the per-dataset values behind that range for the post-RoPE pair $Q_\pi^{h,l}$-$K_\pi^{h,l}$ and for $V^{h,l}$. In every model and dataset, the CV lies between 49.6% and 59.2% for the Q_π-K_π pair and between 41.3% and 45.7% for V, which are the ranges reported in the paper.

Table 2: CV (%) of the maximum singular values across heads for the post-RoPE pair $Q_\pi^{h,l}$-$K_\pi^{h,l}$, per model and dataset.

	Mistral-v0.3 7B	Llama-3.1 8B	Phi-3 14B	Qwen3.5 27B	Yi 34B	Llama-3.1 70B
IMDb	49.76	51.13	52.97	52.1	55.21	57.75
HumanEval	50.04	50.94	52.53	51.8	54.98	57.51
GSM8K	49.9	51.0	52.7	51.9	55.1	57.6
arXiv	49.55	51.21	53.38	52.3	55.73	58.03
Cocktail	49.93	51.07	52.63	52.0	54.95	57.69
RULER-128K	50.1	51.3	52.8	52.2	55.3	57.8
SCBench-MT	49.8	51.1	52.9	52.0	55.0	57.7
PG-19	54.5	52.1	50.7	51.5	54.9	59.2

Table 3: CV (%) of the maximum singular values across heads for $V^{h,l}$, per model and dataset.

	Mistral-v0.3 7B	Llama-3.1 8B	Phi-3 14B	Qwen3.5 27B	Yi 34B	Llama-3.1 70B
IMDb	41.86	43.59	45.37	44.5	44.48	45.16
HumanEval	41.31	43.61	45.69	44.6	44.52	44.95
GSM8K	41.6	43.5	45.1	44.4	44.5	45.0
arXiv	42.37	43.63	44.82	44.2	44.81	45.42
Cocktail	41.44	43.56	44.96	44.3	44.96	45.38
RULER-128K	41.7	43.6	45.0	44.4	44.7	45.3
SCBench-MT	41.5	43.5	45.2	44.3	44.6	45.1
PG-19	43.55	42.92	44.73	43.8	45.25	45.58

3 Integration with vLLM (Paper §5.1)

The paper (§5.1) lists what FastDC changes in vLLM [5], namely the attention module, the width of each head's record in a Key-Value (KV) block, and the attention kernels, and what it leaves stock. This section describes the block layout and the handling of padding. Inside a KV block, each KV head occupies a contiguous tile at its own compressed width, and a per-layer offset table locates the tile of each head. Compression is per token. A prefill chunk compresses its own tokens and attends to the compressed K and V of earlier chunks. A prefix-cache block depends only on its token prefix and is reused without recompression. Speculative verification uses the prefill path. Padding to the tensor-core tile size happens only inside the kernel, and the cache stores the unpadded widths.

4 Overall KV Compression Rates at the 99% Accuracy Threshold (Paper §5.3)

Figure 9 of the paper plots the compression rates of six methods at the 90%–99% accuracy thresholds for eight of the 48 model-dataset pairs. Table 4 gives the overall KV compression rate at the 99% threshold for all 48 pairs, that is, for the six models of the paper's Table 2 on the eight datasets of its Table 1. The methods are those of the paper (§5.2). KF-F and KF-P apply Keyformer [1] eviction and then FastDC or Palu [2], with the two techniques contributing equally to the overall rate. KVQ-F and KVQ-P apply 4-bit KVQuant [4] after FastDC or Palu. In every cell, FastDC exceeds Palu by 0.004–0.100. KF-F exceeds KF-P in 47 of the 48 cells, by up to 0.080, and trails by 0.020 in one. KVQ-F exceeds KVQ-P in 43 cells, by at most 0.030, trails by at most 0.020 in three, and ties in two.

Table 4: Overall KV compression rate of the six methods at the 99% accuracy threshold for all 48 model-dataset pairs.

	FastDC	Palu	KF-F	KF-P	KVQ-F	KVQ-P
IMDb						
Mistral-v0.3 7B	0.61	0.55	0.71	0.68	0.90	0.88
Llama-3.1 8B	0.53	0.51	0.72	0.69	0.89	0.88
Phi-3 14B	0.58	0.50	0.80	0.75	0.92	0.90
Qwen3.5 27B	0.57	0.52	0.76	0.72	0.91	0.89
Yi 34B	0.55	0.49	0.75	0.71	0.92	0.91
Llama-3.1 70B	0.63	0.57	0.74	0.72	0.91	0.89
HumanEval						
Mistral-v0.3 7B	0.53	0.50	0.73	0.69	0.89	0.89
Llama-3.1 8B	0.54	0.48	0.76	0.72	0.87	0.89
Phi-3 14B	0.54	0.50	0.77	0.72	0.91	0.90
Qwen3.5 27B	0.613	0.59	0.784	0.77	0.907	0.902
Yi 34B	0.59	0.53	0.78	0.73	0.90	0.91
Llama-3.1 70B	0.56	0.52	0.77	0.74	0.887	0.89
GSM8K						
Mistral-v0.3 7B	0.57	0.52	0.76	0.72	0.90	0.89
Llama-3.1 8B	0.55	0.49	0.74	0.70	0.89	0.88
Phi-3 14B	0.60	0.53	0.79	0.74	0.91	0.90
Qwen3.5 27B	0.573	0.55	0.764	0.754	0.882	0.877
Yi 34B	0.58	0.52	0.81	0.76	0.92	0.91
Llama-3.1 70B	0.501	0.48	0.74	0.734	0.85	0.842
arXiv						
Mistral-v0.3 7B	0.64	0.59	0.84	0.81	0.92	0.90
Llama-3.1 8B	0.65	0.55	0.84	0.82	0.90	0.88
Phi-3 14B	0.62	0.55	0.82	0.80	0.91	0.89
Qwen3.5 27B	0.64	0.56	0.85	0.82	0.91	0.90
Yi 34B	0.64	0.57	0.87	0.84	0.91	0.90
Llama-3.1 70B	0.67	0.58	0.86	0.82	0.92	0.90
Cocktail						
Mistral-v0.3 7B	0.58	0.51	0.79	0.75	0.90	0.89
Llama-3.1 8B	0.56	0.50	0.75	0.70	0.91	0.89
Phi-3 14B	0.67	0.59	0.82	0.77	0.92	0.90
Qwen3.5 27B	0.63	0.57	0.83	0.78	0.92	0.90
Yi 34B	0.60	0.55	0.83	0.78	0.92	0.91
Llama-3.1 70B	0.59	0.52	0.84	0.76	0.92	0.89
RULER-128K						
Mistral-v0.3 7B	0.60	0.54	0.78	0.74	0.91	0.90
Llama-3.1 8B	0.58	0.52	0.77	0.73	0.90	0.89
Phi-3 14B	0.63	0.56	0.81	0.76	0.92	0.90
Qwen3.5 27B	0.603	0.593	0.762	0.75	0.903	0.895
Yi 34B	0.61	0.55	0.80	0.77	0.92	0.91
Llama-3.1 70B	0.624	0.62	0.783	0.778	0.906	0.902
SCBench-MT						
Mistral-v0.3 7B	0.59	0.53	0.77	0.73	0.91	0.90
Llama-3.1 8B	0.57	0.51	0.76	0.72	0.90	0.88
Phi-3 14B	0.62	0.55	0.80	0.75	0.91	0.90
Qwen3.5 27B	0.551	0.53	0.788	0.77	0.88	0.866
Yi 34B	0.60	0.54	0.79	0.76	0.91	0.91
Llama-3.1 70B	0.616	0.59	0.793	0.78	0.905	0.897
PG-19						
Mistral-v0.3 7B	0.65	0.58	0.81	0.77	0.92	0.91
Llama-3.1 8B	0.63	0.56	0.83	0.80	0.91	0.89
Phi-3 14B	0.66	0.59	0.79	0.74	0.91	0.90
Qwen3.5 27B	0.66	0.58	0.80	0.77	0.92	0.90
Yi 34B	0.65	0.57	0.78	0.80	0.92	0.91
Llama-3.1 70B	0.67	0.59	0.83	0.79	0.91	0.89

5 Evaluation-Subset Accuracy, Cross-Calibration, and Held-Out Workloads (Paper §5.3)

Each dataset is split at random into a 20% calibration subset and an 80% evaluation subset (paper §4.2); the sweep that builds the F_p table measures accuracy on the calibration subset, and the accuracies below are measured on the evaluation subset. Table 5 gives, for the four models of the paper's latency study (§5.4) and the eight datasets, the accuracy retained at the 99% operating points on the evaluation subsets, as a percentage of the uncompressed accuracy measured on the same subsets. All 32 values lie between 98.6% and 99.1%.

Table 5: Accuracy retained at the 99% operating points on the 80% evaluation subsets (% of the uncompressed accuracy on the same subsets).

	Mistral-v0.3 7B	Phi-3 14B	Qwen3.5 27B	Llama-3.1 70B
IMDb	99.1	99.0	98.9	99.1
HumanEval	98.6	98.6	98.8	98.6
GSM8K	98.6	98.7	98.6	98.6
arXiv	99.0	99.1	99.1	98.8
Cocktail	98.8	98.8	98.9	98.7
RULER-128K	98.9	98.9	98.7	98.8
SCBench-MT	98.7	98.8	98.8	98.9
PG-19	99.0	99.0	99.0	99.1

Table 6 reports cross-calibration for Llama-3.1 70B. Here, R is calibrated on arXiv (the R_a of arXiv in the paper's Table 3) instead of on random tokens, and the other seven datasets are served at their 99% operating points. In the table, Δrate is the change in the compression rate and Δaccuracy the change in the retained accuracy, in points, both relative to the random-token R. The compression rate changes by at most 0.006 and the retained accuracy by at most 0.4 points, which is the cross-calibration result of the paper (§3.2).

Table 6: Cross-calibration on Llama-3.1 70B: changes in the compression rate and the retained accuracy at the 99% operating points when R is calibrated on arXiv instead of on random tokens.

Served dataset	Δrate	Δaccuracy (points)
IMDb	+0.003	−0.1
HumanEval	−0.003	−0.3
GSM8K	−0.006	−0.4
Cocktail	+0.004	−0.2
RULER-128K	−0.005	−0.2
SCBench-MT	−0.004	−0.3
PG-19	+0.005	−0.2

Held-out workloads. Table 7 evaluates the random-token R of Llama-3.1 70B on four held-out workloads outside the eight datasets of the paper. MGSM [9] covers multilingual mathematical reasoning and is scored by answer accuracy. LCC [3] and RepoBench-P [6] cover repository-level code completion and are scored by edit similarity. BFCL [7] covers tool use and is scored by abstract syntax tree accuracy. A multi-turn subset of ShareGPT [8] is scored by ROUGE-2 of the final response. Each workload is split into a 20% calibration subset and an 80% evaluation subset in the same way as the datasets above. The first two rows compare the workload's own R_a with R_r by the difference ratio and the largest principal angle of the paper's Table 3. The next three rows give the compression rates of FastDC and of Palu at the 99% operating point and the accuracy that FastDC retains on the evaluation subset at that point. The last two rows repeat the cross-calibration above, with R calibrated on arXiv. The difference ratio stays within 0.20%–0.26% and the largest principal angle within 1.14°. FastDC reaches rates of 0.50–0.63 (Palu 0.48–0.57) and retains 98.6%–98.9% of the uncompressed accuracy, and calibrating on arXiv changes the rates by at most 0.006 and the accuracy by at most 0.3 points. For Llama-3.1 8B, the 99% operating points reach rates of 0.53, 0.54, 0.51, and 0.60 on the same four workloads.

Table 7: Held-out workloads served with the random-token R of Llama-3.1 70B.

	MGSM	LCC+RepoBench-P	BFCL	ShareGPT
Difference ratio	0.24%	0.26%	0.25%	0.20%
Largest principal angle	1.05°	1.14°	1.08°	1.00°
Rate, FastDC	0.50	0.57	0.54	0.63
Rate, Palu	0.48	0.51	0.50	0.57
Retained accuracy (%)	98.7	98.6	98.6	98.9
Δrate, arXiv R	−0.006	−0.005	−0.005	+0.003
Δaccuracy (points), arXiv R	−0.3	−0.3	−0.3	−0.2

6 Latency Variability (Paper §5.4)

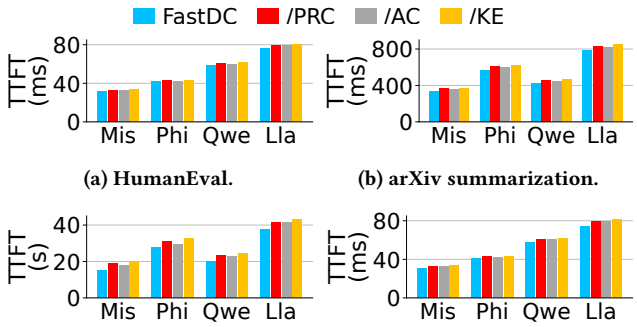
Each latency in the paper is the average over the requests of a trace from one run (paper §5.2). Table 8 quantifies the dispersion behind these averages for FastDC and Palu on Llama-3.1 70B with two measures per dataset. The first is the ratio between the 99th percentile (P99) and the mean of the per-request Time-To-First-Token (TTFT) and Time-Between-Tokens (TBT) within a trace. The second is the standard deviation of the per-run mean TTFT and mean TBT across three runs of the same trace at the same Poisson arrival rate with different seeds, as a percentage of the mean. For FastDC, the P99 TTFT is 1.3–2.7× the mean and the P99 TBT 1.6–1.9× the mean; for Palu, the ratios are 1.6–2.9× and 1.7–2.0×. The FastDC ratios for HumanEval and arXiv are reported jointly. The within-trace spread follows each trace's mix of input and output lengths and the batching of requests, and it is common to all methods, which replay the same trace. Across runs, the standard deviation of the mean is 2%–5% of the mean for both methods.

Table 8: Dispersion of TTFT and TBT for FastDC and Palu on Llama-3.1 70B: P99/mean within a trace, and the standard deviation of the mean over three runs as a percentage of the mean.

	HumanEval	arXiv	RULER-128K	PG-19
P99/mean of TTFT				
FastDC	2.2–2.7		1.3	1.8
Palu	2.5	2.9	1.6	1.9
P99/mean of TBT				
FastDC	1.6	1.6	1.6	1.9
Palu	1.8	1.7	1.7	2.0
Standard deviation of the mean TTFT (%)				
FastDC	4	3	2	3
Palu	5	3	4	3
Standard deviation of the mean TBT (%)				
FastDC	5	4	3	4
Palu	5	4	3	4

7 Ablation Study (Paper §5.4)

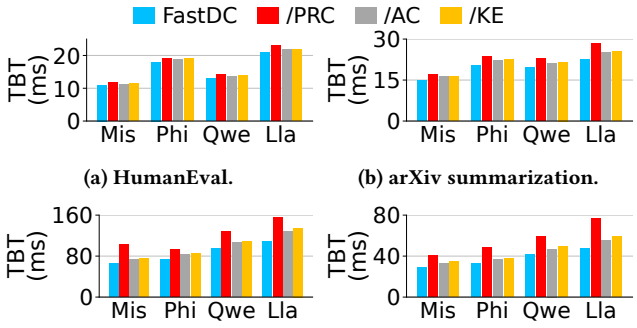
The paper evaluates three variants of FastDC. FastDC/PRC removes Post-RoPE QK Compression; like Palu, it compresses K before RoPE and decompresses K before attention. FastDC/AC removes Adaptive Compression and uses one compression rate for all heads that retains 99% of the uncompressed accuracy. FastDC/KE removes the attention Kernel Enhancement. Figures 1 and 2 show the average TTFT and TBT of FastDC and the three variants on HumanEval, arXiv summarization, RULER-128K, and PG-19 for Mistral-v0.3 7B, Phi-3 14B, Qwen3.5 27B, and Llama-3.1 70B, labeled Mis, Phi, Qwe, and Lla on the horizontal axes. Every variant is slower than FastDC, by 3%–17% in TTFT and 4%–46% in TBT; the paper (§5.4) gives and explains the differences per variant and dataset. FastDC/PRC shows the largest TBT degradation, except on HumanEval, because of its decompression.



(a) HumanEval. (b) arXiv summarization.

(c) RULER-128K. (d) PG-19.

Figure 1: Average TTFT of FastDC and its three ablation variants on four models and four datasets.



(a) HumanEval. (b) arXiv summarization.

(c) RULER-128K. (d) PG-19.

Figure 2: Average TBT of FastDC and its three ablation variants on four models and four datasets.

8 Sensitivity of R to the Number of Calibration Tokens (Paper §5.4)

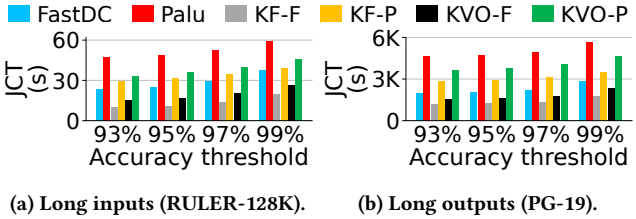
Figure 14(b) of the paper plots, for Llama-3.1 70B on GSM8K, the compression rate and the retained accuracy (called relative accuracy there) against the number of calibration tokens N from which R is computed. Table 9 tabulates the plotted values. For each N, the compression rate is the rate that Equation (1) of the paper assigns at the r of the 99% operating point, and the retained accuracy is the accuracy at that rate as a percentage of the uncompressed accuracy. The paper uses N = 16K. Fewer than 8K tokens raise the rate and lower the accuracy, to 0.80 and 73% with 100 tokens. From 8K tokens on, the rate stays within 0.49–0.51 and the accuracy within 0.3 points of the 16K values.

Table 9: Compression rate and retained accuracy versus the number of calibration tokens N (Llama-3.1 70B, GSM8K).

N	Compression rate	Retained accuracy (%)
100	0.80	73
500	0.64	91
1K	0.56	98.1
2K	0.54	98.6
4K	0.52	98.8
8K	0.509	98.7
16K	0.501	99.0
32K	0.50	99.2
128K	0.49	99.1

9 Job Completion Time under Varying Accuracy Thresholds (Paper §5.4)

Figure 3 shows the average Job Completion Time (JCT) of the six methods of the paper for Llama-3.1 70B at accuracy thresholds of 93%, 95%, 97%, and 99%, on the long-input RULER-128K trace and the long-output PG-19 trace. A lower threshold admits a higher compression rate. As the threshold decreases from 99% to 93%, the JCT of FastDC falls by 31%–38%, versus 16%–20% for Palu, because FastDC further accelerates attention at the higher compression rates. KF-F and KVQ-F follow the same trend.



(a) Long inputs (RULER-128K). (b) Long outputs (PG-19).

Figure 3: Average JCT of the six methods at accuracy thresholds of 93%–99% (Llama-3.1 70B).

10 Time Overhead (Paper §5.4)

The paper measures two sources of time overhead of FastDC: compression, which multiplies R online, and the scheduling in the attention kernel, which assigns attention work to the streaming multiprocessors of the GPU and whose time is estimated from GPU cycle counts. Both are reported as P99 shares of JCT. Tables 10 and 11 give the two shares for Mistral-v0.3 7B, Phi-3 14B, Qwen3.5 27B, and Llama-3.1 70B on arXiv, Cocktail, HumanEval, and PG-19. The P99 compression time is at most 0.7% of JCT. The P99 scheduling

time is at most 3.81% of JCT, and it tends to decrease with longer sequences, because more computation lowers the relative overhead.

Table 10: P99 compression overhead of FastDC as a share of JCT (%).

	arXiv	Cocktail	HumanEval	PG-19
Mistral-v0.3 7B	0.07	0.09	0.3	0.05
Phi-3 14B	0.08	0.04	0.5	0.02
Qwen3.5 27B	0.07	0.01	0.7	0.09
Llama-3.1 70B	0.05	0.08	0.4	0.04

Table 11: P99 kernel scheduling overhead of FastDC as a share of JCT (%).

	arXiv	Cocktail	HumanEval	PG-19
Mistral-v0.3 7B	2.74	2.37	3.81	1.27
Phi-3 14B	2.37	2.27	3.54	0.94
Qwen3.5 27B	1.85	1.64	3.52	0.85
Llama-3.1 70B	2.39	1.98	3.75	0.99

11 Overhead Breakdown per Serving Scenario (Paper §5.4)

Table 5 of the paper summarizes three serving scenarios for Llama-3.1 70B at the 99% operating points. Table 12 gives the full component shares behind it. The JCT of each method is split into the prefill and decode phases, each phase into decompression, attention, and the remaining work, and, in the disaggregated scenario, the KV transfer between the prefill and decode instances. Each scenario is a separate run, so the shares are of each method's own JCT and the JCT ratio compares the two methods within a scenario. In the long-output scenario, prefill is 1.6% of the JCT of Palu and 2% of that of FastDC and is not split further, so its value spans the three prefill rows. Palu spends 6.6%, 28.0%, and 8.5% of its JCT on decompression in the three scenarios, which FastDC removes entirely, and the JCT of FastDC is 0.74, 0.39, and 0.72 of Palu's. For RULER-128K, the P99 compression overhead of FastDC is 0.04% of JCT and the P99 scheduling overhead 2.2%.

Table 12: Component shares of the JCT of each method (%) in three serving scenarios, and the JCT of FastDC relative to Palu (Llama-3.1 70B, 99% operating points).

	Long input (RULER-128K)		Long output (PG-19)		Disaggregated (RULER-128K)	
	Palu	FastDC	Palu	FastDC	Palu	FastDC
Prefill						
Other	38.0	50			24.7	33.3
Decompression	1.1	0	1.6	2	0.8	0
Attention	47.9	39			30.5	25.2
Decode						
Other	0.8	5	14.4	37	6.1	8.1
Decompression	5.5	0	28	0	7.7	0
Attention	6.7	6	56	61	12.2	11.4
KV transfer	–	–	–	–	18.0	22.0
JCT relative to Palu	1.00	0.74	1.00	0.39	1.00	0.72

12 Per-Token and Per-Row Approximation Error (Paper §5.4)

Equation (2) of the paper writes the error of the compressed logits as $(Q_\pi R_\perp)(K_\pi R_\perp)^T$, where $R_\perp$ holds the discarded columns of R, and Equation (3) bounds its aggregate Frobenius norm relative to $Q_\pi K_\pi^T$ by $r/(1-r)$. Here, r is the matrix energy removal rate (paper §4.2). The paper (§5.4) reports per-token and per-row statistics for Llama-3.1 70B at the 99% operating points of the eight datasets, where $r \le 0.039$; this section adds the absolute changes. Per token, the discarded share of energy, $\|(Q_\pi R_\perp)_i\|^2 / \|(Q_\pi)_i\|^2$ for row i and likewise for K_π, has a median of $1.4r$, a P99 of $3.7r$, and a maximum of $10r$ over the tokens of all eight datasets, and no token loses more than half of its energy. Per row, the relative perturbation of a logit row has a P99 of 7% and a maximum of 24%, above the aggregate bound, which does not constrain individual rows. The largest absolute change of any logit in a row has a P99 of 0.55 over rows and a maximum of 1.5.

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